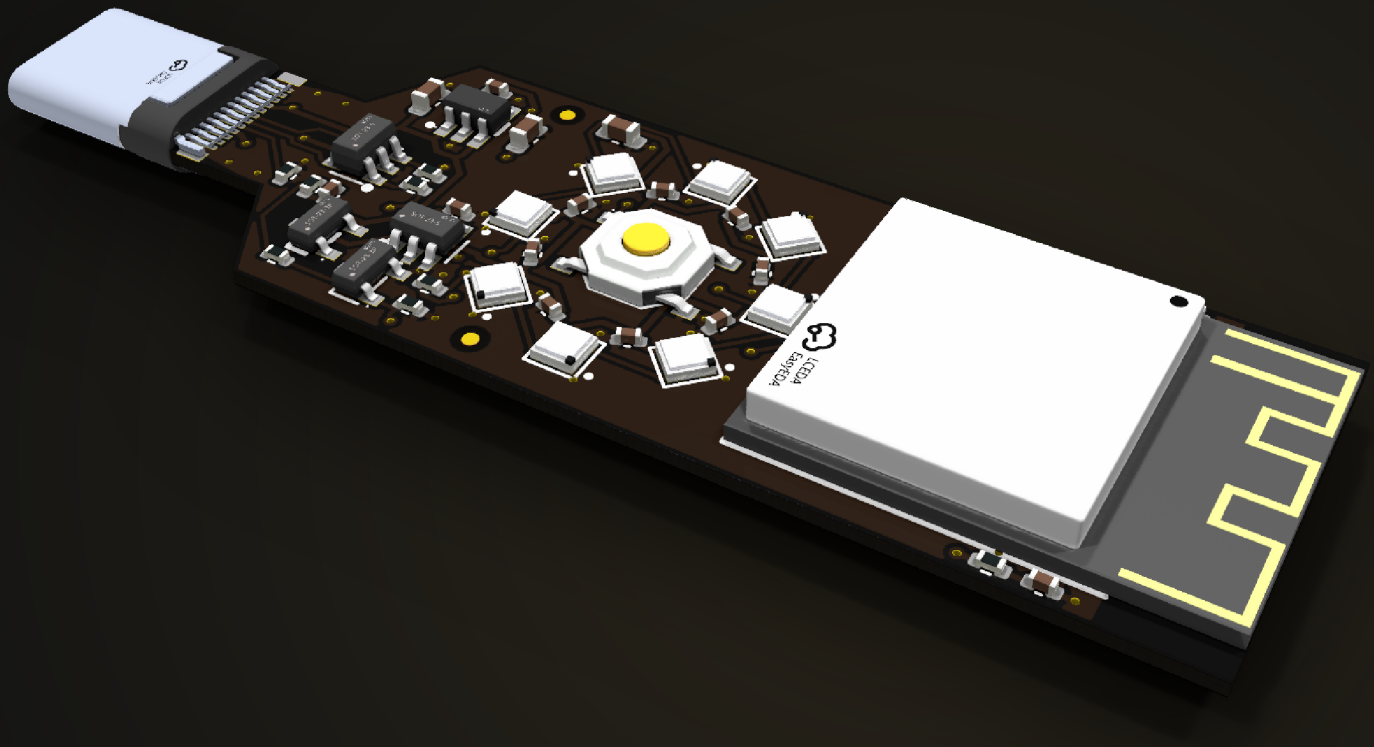
KEYRA · OPEN HARDWARE

Rev 1.0 · designed and fully checked · not manufactured yet

Keyra Stick

Keyra's own board: a hardware password vault the size of a USB stick.



53.1 × 18.5 mm
board, plus the USB-C plug

3.2 mm
at its thickest point

8 lights
RGB, round one button

≈ US$6.7
parts per board

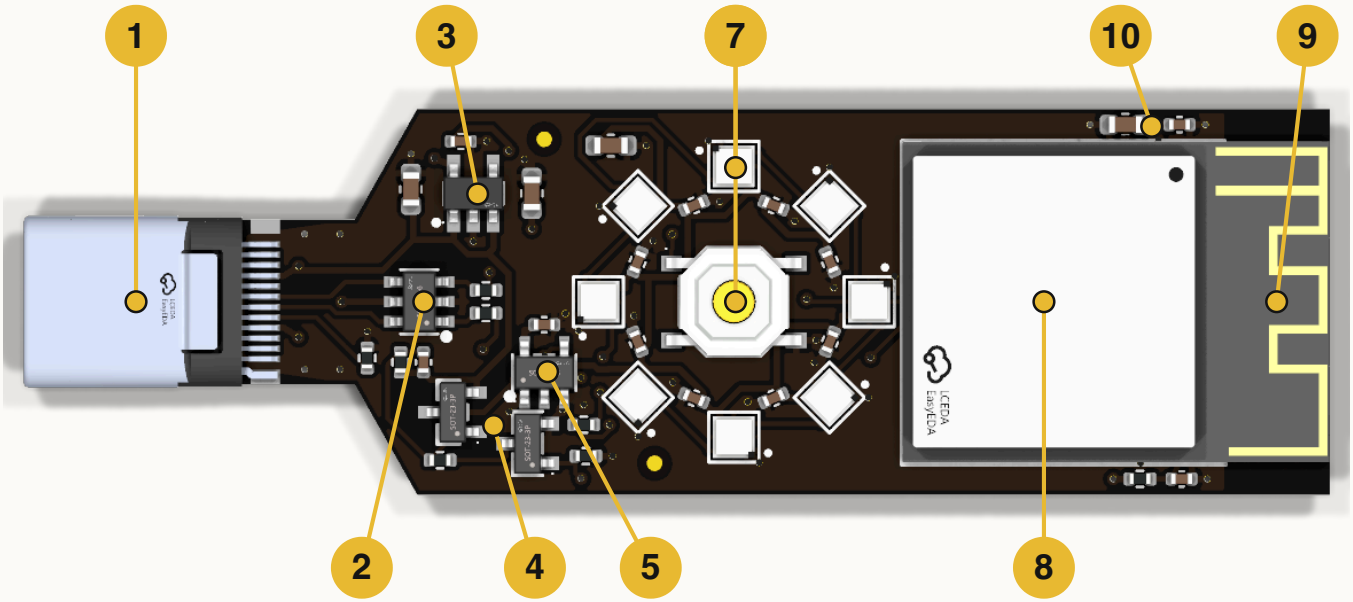
INSIDE THIS GUIDE

1 On the board **2** Size and build **3** How it works **4** Schematic **5** Parts list **6** Ordering from JLCPCB **7** First power-up

Open hardware, MIT licence · github.com/hasanalaaa/keyra · hardware/pcb/keyra-stick

On the board

Every part sits on the top, so the factory assembles it in one pass. The plug end goes into the computer; the antenna end stays outside.



1 USB-C plug J1

Plugs straight into the computer. The board's 0.8 mm tongue sits inside the plug.

2 Static protection U2

Catches static from the port before it reaches anything else.

3 3.3 V regulator U3

Turns the port's 5 V into the module's 3.3 V, up to 600 mA.

4 Light power switch Q1, Q2

The firmware turns the ring's 5 V on and off (IO7).

5 Light signal booster U4

Lifts the chip's 3.3 V signal to the 5 V swing the lights expect.

6 Confirm button SW1

Every sensitive action waits for a press here (IO5).

7 Ring of eight lights D1 to D8

One status colour, seen from any side (IO6).

8 ESP32-S3 module U1

Processor, 8 MB flash, Wi-Fi and Bluetooth. The vault lives here, encrypted.

9 Antenna

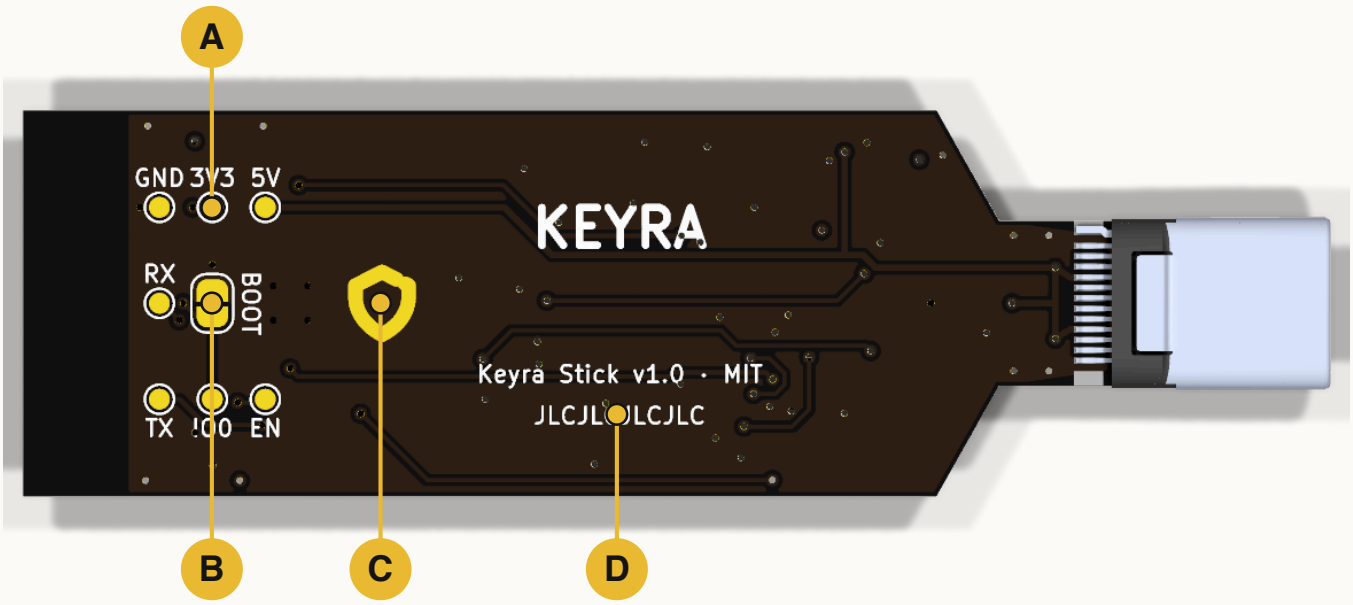
No copper under it on any layer. Keep metal away from this end.

10 Module support C4, C17, R4, C5

Supply capacitors and the reset delay, right at the module's pins.

The back

No parts on the back: bare gold pads for bring-up and recovery, and the marks.



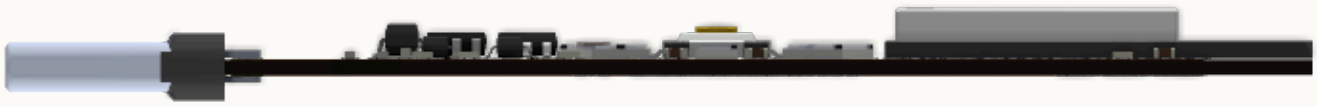
A **Test pads** TP1 to TP7
5V, 3V3, GND, EN, IO0, TX and RX on a 2.54 mm grid, for probes or a pogo-pin jig.

B **BOOT pads** JP1
Bridge them with tweezers while plugging in: download mode, for recovery only.

C **Keyra mark**
The gold Contour, plated like the pads.

D **Name and order number**
JLCPCB prints its order number over JLCJLCJLCJLC.

From the side

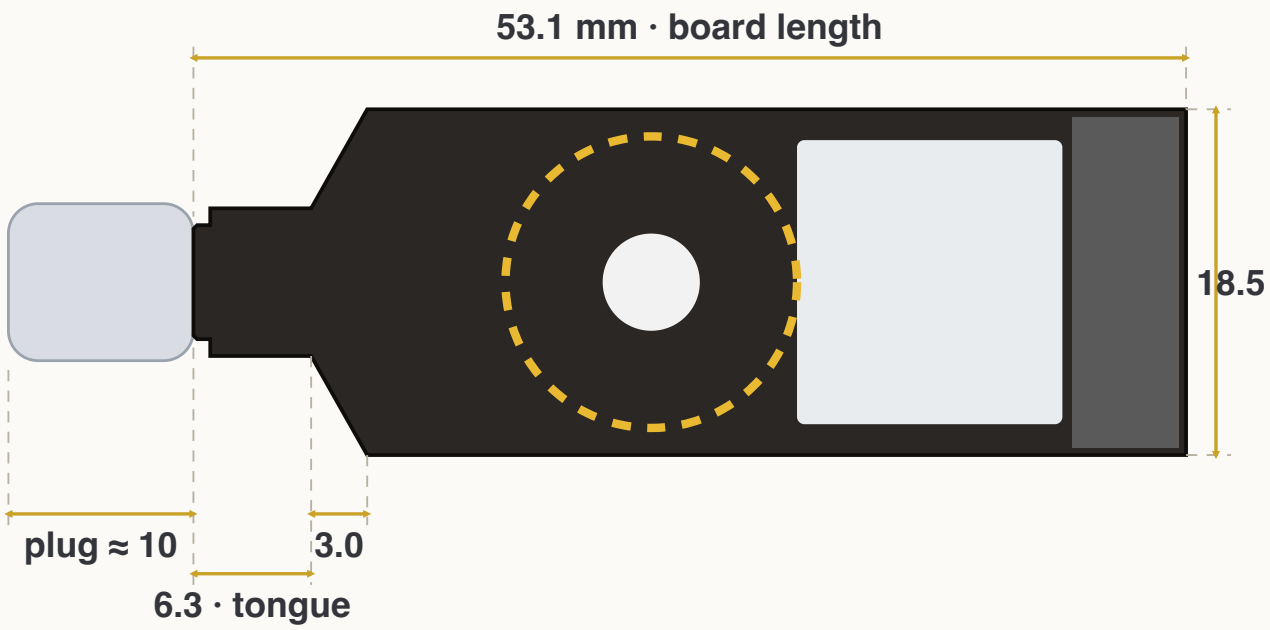


3.2 mm at the module, the thickest point

0.8 mm the board itself

Size and build

The plug decides the board: it clamps a 0.8 mm board, and the tongue follows the plug's drawing (DWG001-A401). Everything else is as narrow as the module allows. Sizes in mm.



Four copper layers



Top parts and signals

Inner 1 solid ground

Inner 2 3.3 V

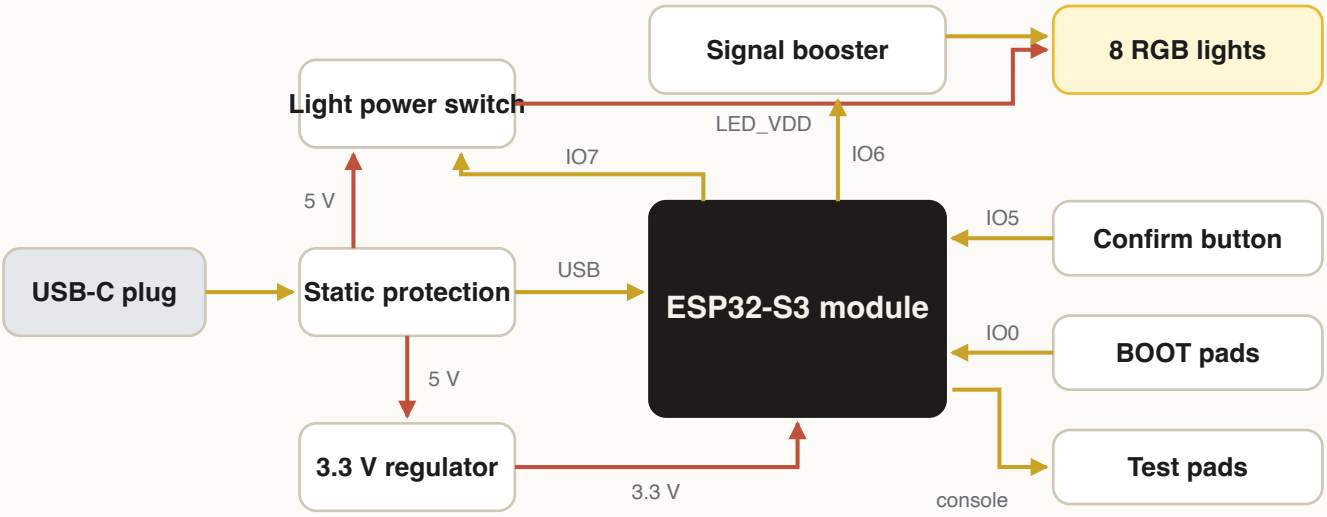
Bottom signals, ground, test pads

The board

Layers	4
Thickness	0.8 mm, as the plug needs
Finish	ENIG (gold)
Colour	black, white print
Assembly	top side, by JLCPCB
Antenna end	no copper on any layer

How it works

Five small blocks around one module. Red arrows carry power, gold ones signals.



Power

The computer's port gives 5 V. The regulator makes 3.3 V for the module; the lights take the 5 V directly, through a switch the firmware controls, so they draw nothing until Keyra runs.

USB

The ESP32-S3 has USB built in: Keyra appears as a keyboard and a security key with no extra chip. Static protection and two small resistors are all the line needs.

Lights

All eight lights show the same colour. They want a 5 V signal, so a one-gate booster turns the chip's 3.3 V signal into 5 V.

Button

The button is on a pin the chip never reads at start-up, so pressing it can never change how the board starts.

Recovery

If the firmware ever needs reinstalling by cable, bridge the BOOT pads while plugging in.

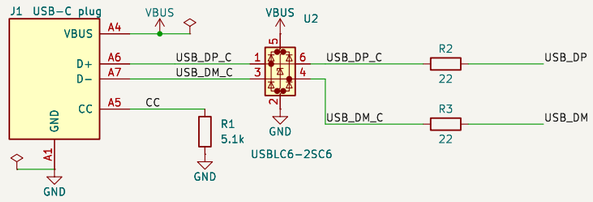
Updates

After the first install, new versions arrive over the air, signed, like on every Keyra board.

What each pin does

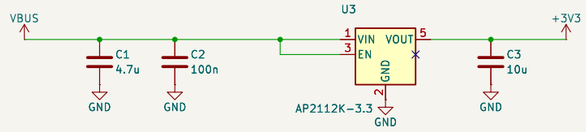
`IO5`	Confirm button (pressed = low)
`IO6`	Light data, through U4
`IO7`	Light power (high = on)
`IO19, IO20`	USB D–, D+
`IO0`	BOOT pads (start-up mode)
`U0TXD, U0RXD`	Console, 115200 baud
`EN`	Reset (low = reset)

1 USB-C plug and protection



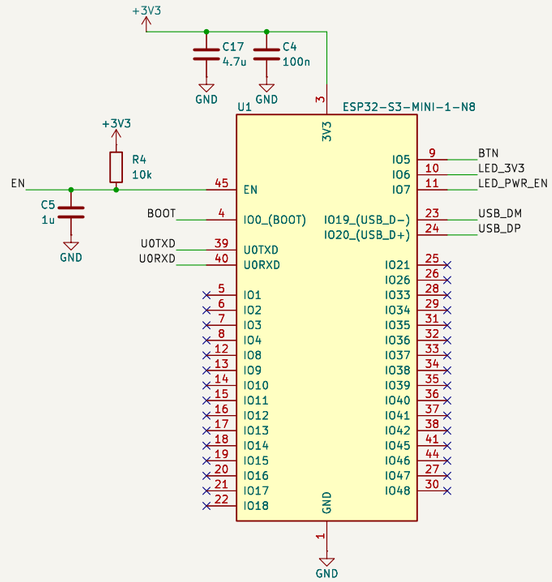
J1 plugs straight into the computer. R1 (5.1 k on CC) asks the host for 5 V.
U2 clamps static on D+, D− and VBUS right behind the plug; R2, R3 (22 R) damp the USB edges.

2 3.3 V supply



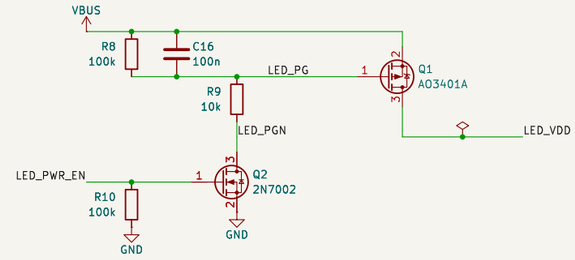
U3: 600 mA LDO for the module (Wi-Fi peaks at about 355 mA). EN tied to VBUS: always on.
C1, C2 at its input, C3 at its output; C17 and C4 sit at the module itself.

3 ESP32-S3 module



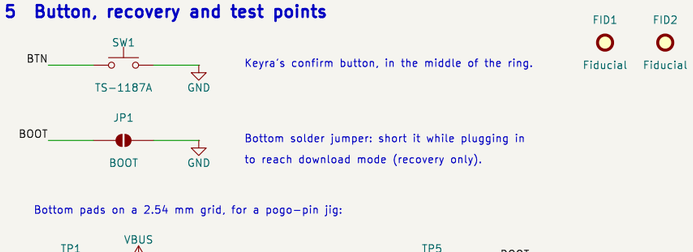
8 MB flash, no PSRAM. The antenna end is the board end: no copper under it.
R4/C5 delay EN at power-up. IO0 keeps the chip's pull-up: JP1 to GND = download mode.
IO19/IO20 are the chip's own USB: keyboard, security key and console.

4 Ring power switch



IO7 high: Q2 pulls Q1's gate down and the ring gets 5 V. R10 keeps it off until the firmware runs.
C16 holds Q1 off while VBUS rises at plug-in; R8 || R9 with C16 make the turn-on about 1 ms.

5 Button, recovery and test points

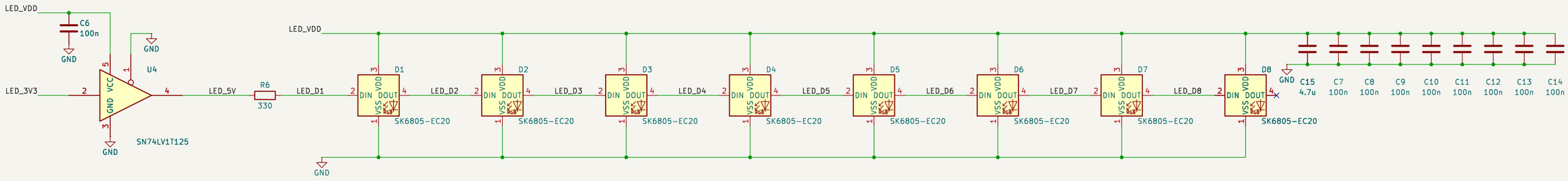


Keyra's confirm button, in the middle of the ring.

Bottom solder jumper: short it while plugging in to reach download mode (recovery only).

Bottom pads on a 2.54 mm grid, for a pogo-pin jig:

6 Status ring



U4 lifts IO6 (3.3 V) to a 5 V data swing; R6 damps the line. Eight SK6805-EC20 in one chain, all showing the same colour; D8's DOUT stays open.
One 100 nF at each LED (C7 to C14) and 4.7 uF bulk (C15), all on the switched LED_VDD.

Notes

Keyra Stick rev 1.0, Keyra's own board: 4 layers, 0.8 mm, 53.1 x 18.5 mm, ENIG. JLCPCB assembles it from fab/ (BOM, CPL).

Every value, net and position comes from gen/design.py; gen/build.sh redraws this sheet and the board and checks both.

Firmware: the keyra-stick board profile (button IO5, eight LEDs on IO6, their supply on IO7).

Generated by gen/make_sch.py from gen/design.py: edit those, not this file
USB-C stick: ESP32-S3-MINI-1-N8, one button, a ring of eight RGB LEDs
Keyra open hardware (MIT)

Sheet: /		
File: keyra-stick.kicad_sch		
Title: Keyra Stick		
Size: A3	Date: 2026-11-03	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 1/1

5 · PARTS LIST

Parts list · 41

JLCPCB assembles every part below from its own stock; the LCSC number names the exact part.

REF	PART	VALUE	PACKAGE	LCSC	WHAT IT DOES
J1	USB-C plug		USB-C	C3151751	USB-C plug
U1	ESP32-S3-MINI-1-N8		15.4 × 20.5 mm	C2913206	The processor module
U2	USBLC6-2SC6		SOT-23-6	C7519	Static protection on USB
U3	AP2112K-3.3		SOT-23-5	C51118	3.3 V regulator
U4	SN74LV1T125		SOT-23-5	C473338	Light signal booster
Q1	AO3401A		SOT-23	C15127	Light power switch
Q2	2N7002		SOT-23	C8545	Drives the light switch
SW1	TS-1187A		5.1 × 5.1 mm	C318884	Confirm button
D1, D2, D3, D4, D5, D6, D7, D8	SK6805-EC20		2 × 2 mm	C2890036	RGB lights, in a ring
R2, R3	Resistor	22	0402	C25092	Damp the USB lines
R4, R9	Resistor	10k	0402	C25744	Reset pull-up; switch gate
R8, R10	Resistor	100k	0402	C25741	Hold the light switch off
R1	Resistor	5.1k	0402	C25905	Asks the computer for 5 V (CC)
R6	Resistor	330	0402	C25104	Damps the light data
C2, C4, C6, C7, C8, C9, C10, C11, C12, C13, C14, C16	Capacitor	100n	0402	C1525	Decoupling at every chip and light
C1, C15, C17	Capacitor	4.7u	0603	C19666	Bulk supply capacitors
C3	Capacitor	10u	0603	C19702	Regulator output
C5	Capacitor	1u	0402	C52923	Reset delay

> Parts cost about US$6.7 per board at single-piece LCSC prices (November 2026): the module about 4.75, the eight lights 0.88, everything else under 1. JLCPCB adds the board, assembly, and a one-off fee per "extended" part type; its quote shows the real total.

Ordering from JLCPCB

Three files from fab/ are all the factory needs. Anyone can order, and it arrives fully assembled.

`keyra-stick-gerbers.zip`
the board itself

`bom.csv`
the parts

`cpl.csv`
where each part goes

1. Open jlcpcb.com → Order now → Add gerber file, and choose keyra-stick-gerbers.zip.
2. Set the board options as in the table below.
3. Turn on PCB Assembly: Standard, top side.
4. Upload bom.csv as the BOM and cpl.csv as the placement file. Every line should match a part in stock.
5. Paste the remark below into the order notes.
6. Check the placement preview (list below), then order.

OPTION	CHOOSE	WHY
Layers	**4**	ground and 3.3 V planes
PCB thickness	**0.8 mm**	the plug clamps exactly this
Surface finish	**ENIG**	flat pads for fine pins; the gold mark
PCB colour	**Black**	any colour works
Impedance control	**No**	USB full speed over a few centimetres
Mark on PCB	**Specify a location**	lands on JLCJLCJLCJLC on the back
Assembly	**Standard, top side**	the plug is a "Standard only" part

Remark for the order

```
J1 straddles the 0.8 mm edge: please solder its bottom-row pads too if possible. Edge rails on
the long sides only, nothing at the USB-C end (the plug overhangs it).
```

In the placement preview

- The plug's pins sit on the pads at the tip.
- The module's pin-1 dot is at its corner nearest the antenna end.
- All eight lights turn the same way round the ring.

First power-up

1. Look at the plug under a light: no solder bridges between its pins.
2. Bridge the BOOT pads on the back with tweezers, plug the stick in, then let go. The chip's USB download port appears.
3. Install Keyra from the browser installer (hasanalaaa.github.io/keyra), choosing Keyra Stick; or flash keyra-firmware-keyra-stick.bin from a release with esptool (offsets in its flash-offsets.txt).
4. Unplug and plug in again. The ring breathes amber: continue with Keyra's normal setup.
5. Later updates arrive over the air, signed, like on any other board.

If something is off

Nothing appears on USB	Check the plug for solder bridges; bridge the BOOT pads to reach download mode.
The ring stays dark	The ring is powered only once Keyra runs. Check the keyra-stick image is installed.
Weak Wi-Fi	Keep the antenna end clear of metal, cases and other plugs.

Changing the design

Every part, net and position lives in gen/design.py. gen/build.sh redraws the schematic and the board from it, routes the board, checks both (zero errors, schematic and board identical), and rewrites every file in fab/, the renders and this guide.